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Fourier transform representation of the generalized hypergeometric functions with applications to the confluent and Gauss hypergeometric functions

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ABSTRACT

We present a Fourier transform representation of the generalized hypergeometric functions. We then use this representation to evaluate integrals of products of two generalized hypergeometric functions. A number of integral identities for confluent and Gauss hypergeometric functions are presented. The results for Euler's gamma function are deduced as special cases.

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1. Introduction

The generalized hypergeometric function is defined by [4, p. 155]

$$ {}_pF_q(a_1,\dots,a_p;b_1,\dots,b_q;z) := \sum_{n=0}^{\infty} \frac{(z)^n (a_1)_n \cdots (a_p)_n}{n!\,(b_1)_n \cdots (b_q)_n}, \tag{1} $$

where $(\lambda)_n := \frac{\Gamma(\lambda+n)}{\Gamma(\lambda)}$ is the Pochhammer symbol. The series in (1) converges absolutely for all z if $p \le q$ and for $|z| < 1$ if $p = q + 1$, and diverges for all $z \neq 0$ if $p > q + 1$. Many functions that arise in physics and engineering are special cases of generalized hypergeometric functions.

The generalized hypergeometric functions have the following integral representations [4, p. 161]:

$$ {}_pF_q(a_1,\dots,a_p;b_1,\dots,b_q;z) = \frac{1}{\Gamma(a_p)} \int_0^{\infty} t^{a_p-1} e^{-t}\, {}_{p-1}F_q((a_{p-1});(b_q);zt)dt \quad (Re(a_p) > 0; p \le q+1). \tag{2} $$

We will use (2) to find the Fourier transform (FT) [9, Chapter 12] representation of the generalized hypergeometric functions.

Some of generalized hypergeometric functions have special names. For example, ${}_1F_1(a;b;z)$ also denoted by $\Phi(a;b;z)$ is called a confluent hypergeometric function (CHF) and satisfies the following, Kummer's differential equation [4, p. 287]:

$$ z\frac{d^2u}{dz^2} + [b-z]\frac{du}{dz} - au = 0. $$

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The function ${}_2F_1(a, b; c; z)$ is often called the hypergeometric function or Gauss's hypergeometric function (GHF). ${}_2F_1(a, b; c; z)$ some time denoted by $F(a, b; c; z)$. It satisfies the following hypergeometric differential equation [4. p. 260]:

$$z(1-z)\frac{d^2u}{dz^2} + [c-(a+b+1)z]\frac{du}{dz} - abu = 0.$$

Choosing the parameters a, b, c in ${}_1F_1(a; b; z)$ and ${}_2F_1(a, b; c; z)$, appropriately, most of the special functions of mathematical physics such as the Bessel, Leguerre and Legendre functions can be obtained.

The generalized Kampé de Fériet hypergeometric function of two variables is defined by [6, p. 63]

$$F^{p:q;k}_{l:r;s}\left[\begin{matrix}(a_p): & (b_q); & (c_k); \\ (\alpha_l): & (\beta_r); & (\gamma_s);\end{matrix}\; x, y\right] = \sum_{m,n=0}^{\infty} \frac{\prod_{j=1}^{p}(a_j)_{m+n}\prod_{j=1}^{q}(b_j)_m\prod_{j=1}^{k}(c_j)_n}{\prod_{j=1}^{l}(\alpha_j)_{m+n}\prod_{j=1}^{r}(\beta_j)_m\prod_{j=1}^{s}(\gamma_j)_n}\frac{x^m}{m!}\frac{y^n}{n!}, \tag{3}$$

where (λ_p) denotes the array of p parameters $\lambda_1, \lambda_2, \ldots, \lambda_p$. For convergence,

1. $p+q < l+r+1, p+k < l+s+1, |x| < \infty, |y| < \infty$, or
2. $p+q = l+r+1, p+k = l+s+1$, and

$$\begin{cases} |x|^{1/(p-l)} + |y|^{1/(p-l)} < 1, & \text{if } p > l, \\ \max\{|x|, |y|\} < 1, & \text{if } p \le l. \end{cases}$$

FT and distributional representations of Euler's gamma and the generalized gamma functions are given in [2] and [8] respectively. In [7] Tassaddiq used FT and distributional representations to investigate the Fermi–Dirac and Bose–Einstein functions. Distributional representation is given by an infinite series of Dirac-delta functions and has led to new results. In this paper we obtain the FT representation of the generalized hypergeometric functions and consequently of confluent and Gauss hypergeometric functions. The results for Euler's gamma function, obtained earlier in [2], are deduced as special cases. We represent generalized Kampé de Fériet's hypergeometric function as an integral of products of two generalized hypergeometric functions.

The plan of this paper is as follows. The FT representation of the generalized hypergeometric functions is given in Sections 2. In Section 3 we present the main result. A number of integral identities for confluent and Gauss hypergeometric functions are presented in Section 4.

2. Fourier representations of the generalized hypergeometric functions

In this section we obtain Fourier and distributional representations of the generalized hypergeometric functions. Substituting $t = e^y$ in (2), we get the FT representation of the generalized hypergeometric function

$$\begin{aligned}&\Gamma(\sigma+i\tau)_pF_q(\sigma+i\tau, a_1, \ldots, a_{p-1}; b_1, \ldots, b_q; z)\\ &= \sqrt{2\pi}\,\mathcal{F}[e^{\sigma y}\exp(-e^y)_{p-1}F_q(a_1, \ldots, a_{p-1}; b_1, \ldots, b_q; ze^y); \tau] \quad (\sigma > 0; p \le q+1),\end{aligned} \tag{4}$$

where $\mathcal{F}[\varphi; \tau]$ is the FT of φ defined by [9, Chapter 12]

$$\mathcal{F}[\varphi; \tau] := \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{iy\tau}\varphi(y)dy \quad (\tau \in \mathbb{R} > 0).$$

Remark Using the series expansion

$$\begin{aligned}&e^{\sigma y}\exp(-e^y)_{p-1}F_q(a_1, \ldots, a_{p-1}; b_1, \ldots, b_q; ze^y)\\ &= \sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\frac{(z)^n(-1)^m(a_1)_n\cdots(a_{p-1})_n}{n!m!(b_1)_n\cdots(b_q)_n}e^{(\sigma+m+n)y} \quad (\sigma > 0; p \le q+1),\end{aligned} \tag{5}$$

and noting that, the FT of the exponential gives a delta function, [9, p. 253]

$$\mathcal{F}[e^{wy}; \tau] = \sqrt{2\pi}\,\delta(\tau - iw),$$

we can rewrite (4) as a series of delta functions

$$\begin{aligned}&\Gamma(\sigma+i\tau)_pF_q(\sigma+i\tau, a_1, \ldots, a_{p-1}; b_1, \ldots, b_q; z)\\ &= 2\pi\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\frac{(z)^n(-1)^m(a_1)_n\cdots(a_{p-1})_n}{n!m!(b_1)_n\cdots(b_q)_n}\delta(\tau - (\sigma+n+m)i) \quad (\sigma > 0; p \le q+1).\end{aligned} \tag{6}$$

This is the distributional representation of the generalized hypergeometric function. This representation is only meaningful when defined as the inner product with a function that belongs to a space of the required "good functions". For a detailed discussion of good functions see [3].

By putting $z = 0$ in (4) and (6), the FT and the distributional representations of the gamma function obtained earlier in [2] can be deduced as special cases

$$\Gamma(\sigma + i\tau) = \sqrt{2\pi}\mathcal{F}[e^{\sigma y}\exp(-e^y); \tau] \qquad (\sigma > 0), \tag{7}$$

$$\Gamma(\sigma + i\tau) = 2\pi \sum_{m=0}^{\infty} \frac{(-1)^m}{m!}\delta(\tau - (\sigma + m)i) \quad (\sigma > 0). \tag{8}$$

The FT representation of the CHF is given by

$$\Gamma(\sigma + i\tau)_1F_1(\sigma + i\tau; b; z) = \sqrt{2\pi}\mathcal{F}[e^{\sigma y}\exp(-e^y)_0F_1(-; b; ze^y); \tau] \quad (\sigma > 0), \tag{9}$$

and the distributional representation is given by

$$\Gamma(\sigma + i\tau)_1F_1(\sigma + i\tau; b; z) = 2\pi \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} \frac{(z)^n(-1)^m}{n!m!(b)_n}\delta(\tau - (\sigma + n + m)i) \quad (\sigma > 0). \tag{10}$$

Similarly, we obtain the FT and the distributional representations of the GHF:

$$\Gamma(\sigma + i\tau)_2F_1(\sigma + i\tau, b; c; z) = \sqrt{2\pi}\mathcal{F}[e^{\sigma y}\exp(-e^y)_1F_1(b; c; ze^y); \tau] \quad (\sigma > 0). \tag{11}$$

$$\Gamma(\sigma + i\tau)_2F_1(\sigma + i\tau, b; c; z) = 2\pi \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} \frac{(-1)^m(z)^n(b)_n}{m!n!(c)_n}\delta(\tau - (\sigma + n + m)i) \quad (\sigma > 0). \tag{12}$$

3. Main result

In this section, we use Parseval's identity of the FT and the FT representation of the generalized hypergeometric functions to derive the main result. We find that the integral of products of two generalized hypergeometric functions gives generalized Kampé de Fériet's hypergeometric function. We also, present some applications to the CHFs and GHFs.

Let f and g be Fourier transformable functions, then Parseval's identity [9, p. 232] of the FT states that

$$\int_{-\infty}^{\infty} \mathcal{F}[f(y); \tau]\overline{\mathcal{F}[g(y); \tau]}d\tau = \int_{-\infty}^{\infty} f(y)\overline{g(y)}dy. \tag{13}$$

In the following theorem we establish the main result.

Theorem 1.

$$\begin{aligned}&\int_{-\infty}^{\infty} \Gamma(\sigma + i\tau)_{p+1}F_q(\sigma + i\tau, (a_p); (b_q); z)\Gamma(\delta - i\tau)_{r+1}F_s(\delta - i\tau, (c_r); (d_s); w)d\tau \\ &= \frac{\pi}{2^{\sigma+\delta-1}}\Gamma(\sigma + \delta)F^{1:p;r}_{0:q;s}\left[\begin{matrix}\sigma + \delta : & (a_p); & (c_r); & \frac{z}{2}, \frac{w}{2} \\ - : & (b_q); & (d_s); & \end{matrix}\right] \\ &(\sigma + \delta > 0; p \le q; r \le s; |z| < 1 \text{ if } p = q \text{ and } |w| < 1 \text{ if } r = s).\end{aligned} \tag{14}$$

Proof. By using Parseval's identity of the FT (13) and the FT representation of the generalized hypergeometric functions (4) we have

$$\begin{aligned}&\int_{-\infty}^{\infty} \Gamma(\sigma + i\tau)_{p+1}F_q(\sigma + i\tau, (a)_p); (b_q); z)\Gamma(\delta - i\tau)_{r+1}F_s(\delta - i\tau, (c_r); (d_s); w)d\tau \\ &= 2\pi \int_{-\infty}^{\infty} e^{(\sigma+\delta)y}\exp[-2e^y]\,{}_pF_q((a_p); (b_q); ze^y)_rF_s((c_r); (d_s); we^y)dy.\end{aligned} \tag{15}$$

For convenience, let the left-hand side of (15) be denoted by I. By substituting $t = 2e^y$, the integral on the right-hand side of (15) becomes

$$I = \pi 2^{1-\sigma-\delta}\int_0^{\infty} t^{\sigma+\delta-1}e^{-t}\,{}_pF_q\left((a_p); (b_q); \frac{z}{2}t\right)_rF_s\left((c_r); (d_s); \frac{w}{2}t\right)dt.$$

Replacing ${}_rF_s((c_r); (d_s); \frac{w}{2}t)$ by its series representation and interchanging the order of summation and integration, we get

$$I = \pi 2^{1-\sigma-\delta}\sum_{n=0}^{\infty} \frac{\prod_{j=1}^{r}(c_j)_n}{n!\prod_{j=1}^{s}(d_j)_n}\left(\frac{w}{2}\right)^n \int_0^{\infty} t^{\sigma+\delta+n-1}e^{-t}\,{}_pF_q\left((a_p); (b_q); \frac{z}{2}t\right)dt.$$

Making use of (2) we get

$$I = \pi 2^{1-\sigma-\delta}\sum_{n=0}^{\infty} \frac{\prod_{j=1}^{r}(c_j)_n}{n!\prod_{j=1}^{s}(d_j)_n}\left(\frac{w}{2}\right)^n \Gamma(\sigma + \delta + n)_{p+1}F_q\left(\sigma + \delta + n, (a_p); (b_q); \frac{z}{2}\right).$$

Now, again, replacing ${}_{p+1}F_q(\sigma+\delta+n,(a_p);(b_q);\frac{z}{2})$ by its series representation we have

$$I=\pi 2^{1-\sigma-\delta}\sum_{n,m=0}^{\infty}\frac{\prod_{j=1}^{r}(c_j)_n}{n!\prod_{j=1}^{s}(d_j)_n}\Gamma(\sigma+\delta+n)\frac{(\sigma+\delta+n)_m\prod_{j=1}^{p}(a_j)_m}{m!\prod_{j=1}^{q}(b_j)_m}\left(\frac{w}{2}\right)^n\left(\frac{z}{2}\right)^m$$
$$=\pi 2^{1-\sigma-\delta}\Gamma(\sigma+\delta)\sum_{n,m=0}^{\infty}\frac{(\sigma+\delta)_{m+n}}{n!m!}\frac{\prod_{j=1}^{r}(c_j)_n}{\prod_{j=1}^{s}(d_j)_n}\frac{\prod_{j=1}^{p}(a_j)_m}{\prod_{j=1}^{q}(b_j)_m}\left(\frac{w}{2}\right)^n\left(\frac{z}{2}\right)^m.$$

By using the definition of generalized Kampé de Fériet's hypergeometric function (3), we get (14). □

Corollary 2. *The generalized hypergeometric functions satisfy the following integral equation*

$$\int_{-\infty}^{\infty}\Gamma(\sigma+i\tau)_{p+1}F_q(\sigma+i\tau,a_1,\ldots,a_p;b_1,\ldots,b_q;z)\Gamma(\delta-i\tau)d\tau$$
$$=\pi 2^{1-\sigma-\delta}\Gamma(\sigma+\delta)_{p+1}F_q\left(\sigma+\delta,a_1,\ldots,a_p;b_1,\ldots,b_q;\frac{z}{2}\right)\quad(\sigma+\delta>0;p\le q;|z|<1\text{ if }p=q).\tag{16}$$

Proof. The proof of (16) follows from (14) when we take $w=0$. □

Corollary 3.

$$\int_{-\infty}^{\infty}|\Gamma(\sigma+i\tau)|^2\,{}_{p+1}F_q(\sigma+i\tau,a_1,\ldots,a_p;b_1,\ldots,b_q;z)d\tau$$
$$=\pi 2^{1-2\sigma}\Gamma(2\sigma)_{p+1}F_q\left(2\sigma,a_1,\ldots,a_p;b_1,\ldots,b_q;\frac{z}{2}\right)\quad(\sigma>0;p\le q;|z|<1\text{ if }p=q).\tag{17}$$

Proof. Taking $\sigma=\delta$ in (16) we get (17). □

The following integral identities for CHFs and GHFs are consequences of the above theorem:

$$\int_{-\infty}^{\infty}\Gamma(\sigma+i\tau)_1F_1(\sigma+i\tau;b;z)\Gamma(\delta-i\tau)d\tau=\frac{\pi}{2^{\sigma+\delta-1}}\Gamma(\sigma+\delta)_1F_1\left(\sigma+\delta;b;\frac{z}{2}\right)\quad(\sigma,\delta>0),\tag{18}$$

$$\int_{-\infty}^{\infty}|\Gamma(\sigma+i\tau)|^2\,{}_1F_1(\sigma+i\tau;b;z)d\tau=\frac{\pi}{2^{2\sigma-1}}\Gamma(2\sigma)_1F_1(2\sigma;b;\frac{z}{2})\quad(\sigma>0),\tag{19}$$

$$\int_{-\infty}^{\infty}\Gamma(\sigma+i\tau)_2F_1(\sigma+i\tau;b;c;z)\Gamma(\delta-i\tau)d\tau=\frac{\pi}{2^{\sigma+\delta-1}}\Gamma(\sigma+\delta)_2F_1\left(\sigma+\delta;b;c;\frac{z}{2}\right)\quad(\sigma,\delta>0;|z|<1),\tag{20}$$

$$\int_{-\infty}^{\infty}|\Gamma(\sigma+i\tau)|^2\,{}_2F_1(\sigma+i\tau;b;c;z)d\tau=\frac{\pi}{2^{2\sigma-1}}\Gamma(2\sigma)\,{}_2F_1\left(2\sigma;b;c;\frac{z}{2}\right)\quad(\sigma>0;|z|<1).\tag{21}$$

Taking $z=0$, we get the well-known "norm squared" of Γ (see, for example, [5, p143, (5.13.2)]):

$$\int_{-\infty}^{\infty}|\Gamma(\sigma+i\tau)|^2\,d\tau=\pi 2^{1-2\sigma}\Gamma(2\sigma)\quad(\sigma>0).\tag{22}$$

4. Some applications to the CHFs and GHFs

In this section we apply the main result (14) to find some integral identities for CHFs and GHFs.

Theorem 4. *The CHFs satisfy the following identity:*

$$\int_{-\infty}^{\infty}\Gamma(\sigma+i\tau)_1F_1(\sigma+i\tau;b;z)\Gamma(\delta-i\tau)_1F_1(\delta-i\tau;c;z)d\tau$$
$$=\pi 2^{1-\sigma-\delta}\Gamma(\sigma+\delta)_3F_3\left(\sigma+\delta,\frac{b+c-1}{2},\frac{b+c}{2};b,c,b+c-1;2z\right)\quad(\sigma,\delta>0).\tag{23}$$

Proof. Putting $p=r=0$, $q=s=1$ and $w=z$ in (14), we find

$$\int_{-\infty}^{\infty}\Gamma(\sigma+i\tau)_1F_1(\sigma+i\tau;b;z)\Gamma(\delta-i\tau)_1F_1(\delta-i\tau;c;z)d\tau$$
$$=\frac{\pi}{2^{\sigma+\delta-1}}\Gamma(\sigma+\delta)F^{1:0;0}_{0:1;1}\left[\begin{matrix}\sigma+\delta: & -; & -; & \\ -: & b; & c; & \end{matrix}\;\frac{z}{2},\frac{z}{2}\right]\quad(\sigma,\delta>0)\tag{24}$$

Now, using the reduction formula [1, (3.2)]

$$F^{1:0;0}_{0:1;1}\left[\begin{matrix}\alpha: & -; & -; & \\ -: & \nu; & \sigma; & \end{matrix}\;x,x\right]={}_3F_3\left(\alpha,\frac{\nu+\sigma-1}{2},\frac{\nu+\sigma}{2};\nu,\sigma,\nu+\sigma-1;4x\right),$$

we get (23). □

Corollary 5. *The following identities involving the CHFs hold:*

$$\int_{-\infty}^{\infty} \Gamma(\sigma+i\tau)_1F_1(\sigma+i\tau;b;z)\Gamma(\delta-i\tau)_1F_1(\delta-i\tau;b;z)d\tau$$
$$= \pi 2^{1-\sigma-\delta}\Gamma(\sigma+\delta)_2F_2\left(\sigma+\delta, b-\frac{1}{2}; b, 2b-1; 2z\right) \quad (\sigma,\delta>0), \tag{25}$$

$$\int_{-\infty}^{\infty} |\Gamma(\sigma+i\tau)_1F_1(\sigma+i\tau;b;z)|^2\, d\tau = \pi 2^{1-2\sigma}\Gamma(2\sigma)_2F_2\left(2\sigma, b-\frac{1}{2}; b, 2b-1; 2z\right) \quad (\sigma>0). \tag{26}$$

Proof. By taking $b=c$ in (23) we get (25), and as a special case, taking $\sigma=\delta$, we get (26). □

Theorem 6. *The CHFs satisfy the following identity:*

$$\int_{-\infty}^{\infty} \Gamma(\sigma+i\tau)_1F_1(\sigma+i\tau;b;z)\Gamma(\delta-i\tau)_1F_1(\delta-i\tau;b;-z)d\tau$$
$$= \pi 2^{1-\sigma-\delta}\Gamma(\sigma+\delta)_2F_3\left(\frac{\sigma+\delta}{2}, \frac{\sigma+\delta+1}{2}; \frac{b}{2}, \frac{b+1}{2}, b; -\frac{z^2}{4}\right) \quad (\sigma,\delta>0). \tag{27}$$

Proof. Putting $p=r=0$, $q=s=1$, $b=c$ and $w=-z$ in (14), and using the reduction formula [1, (3.7)]

$$F^{1:0;0}_{0:1;1}\left[\begin{matrix}\alpha: & -; & -; \\ -: & \nu; & \nu;\end{matrix}\; x, -x\right] = {}_2F_3\left(\frac{\alpha}{2}, \frac{\alpha+1}{2}; \frac{\nu}{2}, \frac{\nu+1}{2}, \nu; -x^2\right),$$

we obtain the result (27). □

We get similar results for the GHFs.

Theorem 7. *The GHFs satisfy the following identities:*

$$\int_{-\infty}^{\infty} \Gamma(\sigma+i\tau)_2F_1(\sigma+i\tau,b;c;z)\Gamma(\delta-i\tau)_2F_1(\delta-i\tau,b;c;-z)d\tau$$
$$= \pi 2^{1-\sigma-\delta}\Gamma(\sigma+\delta)_4F_3\left(\frac{\sigma+\delta}{2}, \frac{\sigma+\delta+1}{2}, b, c-b; \frac{c}{2}, \frac{c+1}{2}, c; \frac{z^2}{4}\right) \quad (\sigma,\delta>0; Re(c)>Re(b)>0>0; |z|<1), \tag{28}$$

$$\int_{-\infty}^{\infty} \Gamma(\sigma+i\tau)_2F_1(\sigma+i\tau,b;2b;z)\Gamma(\delta-i\tau)_2F_1(\delta-i\tau,c;2c;-z)d\tau$$
$$= \pi 2^{1-\sigma-\delta}\Gamma(\sigma+\delta)_4F_3\left(\frac{\sigma+\delta}{2}, \frac{\sigma+\delta+1}{2}, \frac{b+c}{2}, \frac{b+c+1}{2}; b+\frac{1}{2}, c+\frac{1}{2}, b+c; \frac{z^2}{4}\right) \quad (\sigma,\delta>0; |z|<1). \tag{29}$$

Proof. By using a similar procedure and using the following reduction formulas [1, (3.4), (3.5)]:

$$F^{1:1;1}_{0:1;1}\left[\begin{matrix}\alpha: & \lambda; & \lambda; \\ -: & \nu; & \nu;\end{matrix}\; x, -x\right] = {}_4F_3\left(\frac{\alpha}{2}, \frac{\alpha+1}{2}, \lambda, \lambda-\nu; \nu, \frac{\nu}{2}, \frac{\nu+1}{2}; x^2\right),$$
$$F^{1:1;1}_{0:1;1}\left[\begin{matrix}\alpha: & \lambda; & \mu; \\ -: & 2\lambda; & 2\mu;\end{matrix}\; x, -x\right] = {}_4F_3\left(\frac{\alpha}{2}, \frac{\alpha+1}{2}, \frac{\lambda+\mu}{2}, \frac{\lambda+\mu+1}{2}; \lambda+\frac{1}{2}, \mu+\frac{1}{2}, \lambda+\mu; x^2\right),$$

we obtain (28) and (29). □

We can get more integral formulae by different choices of the parameters a, b and c.

5. Conclusion

In this paper, we obtained the Fourier transform representation of the generalized hypergeometric functions ${}_pF_q((a_p);(b_q);z)$. By using this representation we found that the integral of products of two generalized hypergeometric functions is a generalized Kampé de Fériet's hypergeometric function.

It is worth mentioning here that, whenever a generalized hypergeometric function reduces to the CHF or GHF and other hypergeometric functions, the results become more important for applications. Most of the special functions of mathematical physics and engineering, such as the Bessel, Leguerre and Legendre functions, can be expressed in terms of the CHFs or GHFs. Therefore, the main result can play an important role in the theory of special functions of applied mathematics and mathematical physics.

We conclude our investigation by remarking that several formulae can be obtained from the main result by appropriate choices of the parameters.

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